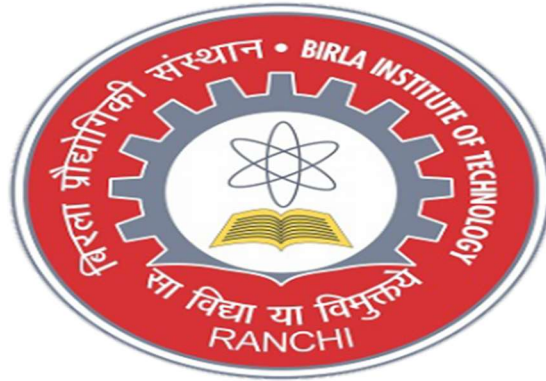




Birla Institute of Technology, Mesra

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“Disk Scheduling”

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— **Harsh Vardhan Sain**

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1. Introduction

In a computer system, the disk is one of the most important components responsible for storing and retrieving data permanently. Unlike the main memory (RAM), which is volatile and loses data when power is turned off, the disk retains data even after shutdown. Since almost every process or application in an operating system requires data from the disk — whether it is opening a file, saving a document, or loading a program — the performance of disk operations has a major effect on the overall speed of the computer.

However, accessing data from the disk is not as fast as accessing data from memory. The disk consists of rotating platters and a movable arm with read/write heads. Every time the operating system needs to read or write data, the disk arm must physically move to the correct track, and the platter must rotate to the right sector. This physical movement takes time, known as seek time and rotational latency. If many processes request disk access at the same time, the disk arm may have to move back and forth repeatedly, increasing the total access time and slowing down the system.

To solve this problem, the operating system uses disk scheduling algorithms. Disk scheduling is the technique of determining the optimal order in which disk I/O requests should be served. The main goal is to reduce the total seek time and improve the response time of the system. By arranging the requests in an intelligent order, disk scheduling ensures that the disk head travels the minimum possible distance, which directly increases the speed and efficiency of the system.

2. Need for Disk Scheduling

When multiple I/O requests arrive, they are stored in a queue. If these requests are handled one by one in the order they come (First-Come, First-Served), the disk arm may move back and forth across the surface many times, causing long delays. Disk scheduling algorithms are needed to minimize such unnecessary movements and to decide the best order for servicing requests.

The main needs for disk scheduling are as follows:

1) To Minimize Seek Time

The most important goal is to reduce seek time. The less the disk arm moves, the faster data can be accessed. For example, if the head is currently at track 50 and requests come for tracks 20, 60, and 90, it is better to serve track 60 first rather than jumping directly to 20 or 90.

2) To Improve Throughput

Throughput means the number of I/O requests completed per unit time. Efficient scheduling ensures more requests are served in less time, improving overall disk performance.

3) To Reduce Response Time

Response time is the time between making a request and getting the result. By scheduling disk operations efficiently, the system can respond faster to user and process requests.

4) *To Ensure Fairness*

Fairness means all processes should get a chance to access the disk. A good scheduling method prevents starvation, where some requests are delayed for too long because others keep getting priority.

5) To Optimize System Performance

Disk scheduling not only improves disk access but also enhances CPU utilization. When the disk performs faster, the CPU does not have to wait for I/O operations, allowing better overall system performance.

3. Structure of Disk

A disk (or hard disk) is a secondary storage device used in computer systems to store large amounts of data permanently. It is a magnetic storage device that can store data even when the power is turned off. Understanding the structure of a disk helps us know how data is stored and accessed efficiently.

1. Physical Structure

A hard disk consists of one or more platters that are circular disks made of metal or glass and coated with a magnetic material. Each platter spins at high speed around a central spindle. The data is stored magnetically on both sides of each platter.

Each platter surface has a read/write head attached to an arm assembly. These heads move across the disk surface to read or write data. All the heads move together as a single unit when the arm moves.

2. Logical Structure

The disk surface is divided into several parts to organize data systematically:

a) Tracks

Each platter surface is divided into many concentric circles known as tracks. Each track acts as a path for storing data. The outer tracks are longer and can store more data than the inner tracks.

b) Sectors

Each track is further divided into small sections called sectors. A sector is the smallest physical storage unit on a disk and typically stores 512 bytes or 4 KB of data.

c) Cylinders

A cylinder is formed by aligning tracks that are directly above and below each other across all platters. When the disk head moves to a specific track on one platter, it is also positioned on the same track of other platters, forming a cylinder. This helps access data quickly across multiple platters without moving the arm much.

d) Clusters and Blocks

In some systems, sectors are grouped together to form clusters or blocks for efficient data storage and retrieval.

4. Types of Disk Scheduling Algorithms

5.1 FCFS (First Come First Serve)

FCFS serves the disk requests in the order they arrive. It is simple but may result in long waiting times when requests are far apart on the disk surface.

5.2 SSTF (Shortest Seek Time First)

SSTF selects the request closest to the current head position. It reduces seek time but can cause starvation for distant requests.

5.3 SCAN (Elevator Algorithm)

In SCAN, the disk arm moves in one direction serving requests until it reaches the end, then reverses direction. It behaves like an elevator moving up and down, hence the name.

5.4 C-SCAN (Circular SCAN)

C-SCAN moves the head in one direction only. After reaching the end, it quickly returns to the beginning without serving any requests. It provides uniform wait time compared to SCAN.

5.5 LOOK

LOOK works like SCAN but instead of going till the end of the disk, it reverses direction when there are no more requests ahead.

5.6 C-LOOK

C-LOOK is a modified version of C-SCAN that only goes as far as the last request in each direction, then jumps back to the first request.

6. Comparison Table of Algorithms

Algorithm	Average Seek Time	Starvation Possibility	Remarks
FCFS	High	No	Simple but inefficient for heavy load.
SSTF	Moderate	Yes	Improves performance but may starve requests.
SCAN	Moderate	Low	Balances seek time; fair to all requests.
C-SCAN	Moderate	Low	Uniform wait time for all requests.
LOOK	Low	Low	Efficient and fair scheduling.
C-LOOK	Low	Low	Best for systems with high load.

The table above compares different disk scheduling algorithms based on their performance, starvation, and general remarks. LOOK and C-LOOK usually provide the best performance among them.

7. Advantages and Disadvantages

Disk scheduling is an important technique used by the operating system to manage multiple I/O requests efficiently. It determines the order in which disk read or write requests should be processed. While it offers many benefits, it also has a few limitations depending on the algorithm used.

Advantages

1. Improved Performance:

Disk scheduling reduces the total seek time by organizing requests in an optimal order. This helps in faster data access and improves overall system performance.

2. Better Throughput:

Since more requests can be served in less time, the overall throughput (number of completed I/O operations per second) increases.

3. Reduced Response Time:

Disk scheduling helps in minimizing the average waiting and response time of processes, which improves user experience and system efficiency.

4. Efficient Resource Utilization:

It ensures that the CPU and disk are not left idle for long periods, improving the utilization of system resources.

5. Fairness:

Most disk scheduling algorithms ensure that all processes get a fair chance of accessing the disk, preventing starvation.

Disadvantages

1. **Complex Implementation:**

Some algorithms like SCAN or C-LOOK are more complex to implement compared to simple methods like FCFS.

2. **Possibility of Starvation:**

In algorithms such as SSTF, requests far from the current head position may wait indefinitely if nearby requests keep arriving.

3. **Algorithm Dependence:**

Performance depends heavily on the chosen algorithm and the pattern of I/O requests.

8. Real Life Applications

Disk scheduling plays an important role in the performance of computer systems and is widely used in real-life applications where large amounts of data are stored and accessed frequently.

One of the most common applications is in operating systems. Every time multiple programs or users request data simultaneously, the operating system uses disk scheduling algorithms to decide which request to handle first. This ensures smooth and efficient data access, reducing waiting time and improving overall system speed.

In databases and file servers, disk scheduling helps manage large volumes of read and write operations efficiently. It reduces seek time and ensures that data retrieval is quick, which is crucial for applications like banking systems, e-commerce websites, and online transaction platforms.

Cloud storage systems and data centers also rely on disk scheduling to handle thousands of requests from users around the world. Proper scheduling ensures high throughput and balanced resource usage.

In multimedia systems, such as video streaming platforms, disk scheduling ensures continuous data flow without buffering or delay.

Overall, disk scheduling is essential in any system that requires fast, reliable, and continuous access to stored data.

9. Conclusion

Disk scheduling is a crucial part of operating system design that directly impacts the speed and efficiency of data access. Since a disk drive handles multiple read and write requests from different processes at the same time, it becomes essential to determine the most efficient order in which these requests are executed. Proper disk scheduling reduces seek time, response time, and rotational delay, leading to faster data retrieval and improved system performance.

Various disk scheduling algorithms like **FCFS**, **SSTF**, **SCAN**, **C-SCAN**, **LOOK**, and **C-LOOK** are designed to optimize performance based on different situations. Each has its own advantages and limitations, but their common goal is to minimize disk head movement and maximize throughput.

In modern systems such as servers, databases, and cloud storage, disk scheduling ensures that multiple users and processes can access data efficiently without long waiting times. It not only improves hardware utilization but also enhances the overall reliability and speed of the computer system.

In conclusion, disk scheduling is essential for achieving balanced, fair, and high-performance data access, making it a fundamental aspect of every operating system's resource management.

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